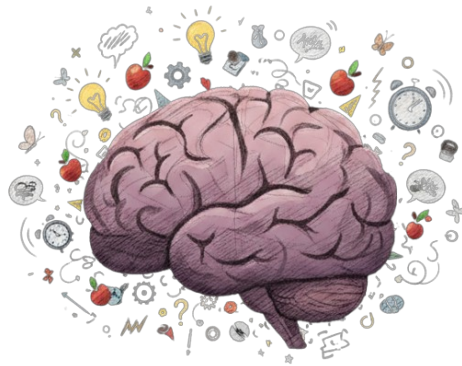


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
SUMMER 2025**



**TEAM MINDMEAL
MINDMEAL**

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REVISION HISTORY

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1 PROBLEM STATEMENT

People with ADHD often struggle to maintain regular eating habits due to challenges in managing time, executive function, and memory. This can lead to skipped meals, poor nutrition, and negative impacts on mental and physical health. Traditional reminder apps are not tailored to the specific cognitive needs of individuals with ADHD, lacking adaptive support, personalization, and accessibility.

There is a critical need for a supportive tool that not only reminds users to eat, but also provides ADHD-friendly features, such as structured routines, gamified motivation, and intelligent meal suggestions tailored to dietary needs or medical conditions.

2 METHODOLOGY

We are developing an AI-assisted mobile app designed to support individuals with ADHD in building healthier eating and daily living habits. The app will recommend meals based on available leftovers or groceries, track user routines, and use AI to suggest daily activities tailored to ADHD behavior patterns. By combining intelligent food suggestions with routine guidance, the app will foster structure, reduce decision fatigue, and promote consistent self-care.

3 VALUE PROPOSITION

Sponsors will benefit from supporting a practical, socially impactful app that addresses real-life challenges faced by individuals with ADHD. By investing in this project, they contribute to the development of an accessible tool that promotes healthier eating through barcode scanning and instant feedback on ingredients, calories, nutrients, and allergens. This project also serves as a valuable academic and research opportunity for instructors and the university, highlighting UTA's commitment to inclusive technology, health awareness, and student innovation.

4 DEVELOPMENT MILESTONES

This list of core project milestones should include all major documents, demonstration of major project features, and associated deadlines. Any date that has not yet been officially scheduled at the time of preparing this document may be listed by month.

Provide a list of milestones and completion dates in the following format:

- Project Charter first draft - July 2025
- System Requirements Specification - July 2025
- Architectural Design Specification - July 2025
- Demonstration of AI meal suggestion feature - July 2025
- Detailed Design Specification - August 2025
- Demonstration of routine tracking feature - August 2025
- Demonstration of barcode scanner to find ingredients - September 2025
- CoE Innovation Day poster presentation - September 2025
- Demonstration of groceries and leftover scanning feature - October 2025
- Final Project Demonstration - November 2025

5 BACKGROUND

ADHD impacts an individuals' attention, focus, memory and executive function. The result is many individuals with ADHD can struggle with object permanence. For instance, if food can't be seen, it is mentally unavailable. This mental unavailability leads to expired leftovers, poor nutrition, and, food waste. While many food tracking or food planning apps are available, most of them are designed for a neurotypical audience, with app designs that require users to check-in regularly or look at many data fields that require high cognitive input - both are challenging for someone with ADHD.

MindMeal intends to fill this gap with an interface and assistant that not only tracks food availability and expiration, but is designed to learn from its users. The app will suggest routines, nudges, and context-sensitive reminders to assist users in adopting a more consistently healthy eating routine. We believe that all of these components in the unmonetized app will foster better nutrition, reduce food waste and improve users' lives for a neurodiverse audience.

6 RELATED WORK

Several commercial apps aim to solve food tracking or meal suggestion problems, but most fall short in ADHD-friendly usability.

Supercook allows users to find recipes based on ingredients, but fails to consider expiration dates or usability concerns. KitchenPal is a pantry manager that suffers from performance and interface issues where users report frequent crashes and confusion. NoWaste and Fridgely provide expiration tracking but rely heavily on tedious manual input, which is a significant barrier for users with ADHD.

IoT-powered solutions like Samsung Family Hub smart fridges offer camera-based tracking, but their cost and complexity limit accessibility. MindMeal distinguishes itself with a lightweight, mobile-first design, image-based inventory, and AI-enhanced routine tracking specifically engineered for ADHD usability.

7 SYSTEM OVERVIEW

MindMeal is a mobile app designed to help individuals with ADHD make healthier food choices and build consistent routines. The system has four main components: an ADHD-friendly User Interface, a Food Scanner Module for barcode and image-based food tracking, a Routine Tracker to learn user habits, and an AI Recommendation Engine for personalized meal and reminder suggestions.

Users scan groceries or leftovers to get ingredient, calorie, and allergen information. The app tracks daily routines and sends well-timed nudges to help users stay on track. Based on available food and behavior patterns, the AI suggests meals and daily structure, reducing food waste and supporting better eating habits with minimal cognitive load.

8 ROLES & RESPONSIBILITIES

STAKEHOLDERS

Primary Stakeholders:

- **End Users:** Adults and students with ADHD, neurodivergent individuals, and anyone seeking to reduce food waste and improve meal habits.
- **Healthcare Professionals:** Dietitians, therapists, and ADHD specialists who may recommend the app or provide feedback on its effectiveness.
- **Educational Institutions:** Schools and universities interested in supporting students with ADHD.

- **Project Sponsor:** [Sponsor Organization/Professor/Client Name], who provides oversight, resources, and final approval.

Point of Contact (Sponsor/Customer Side):

- **[Sponsor/Professor Name, Title, Email]:** Responsible for providing requirements, feedback, and milestone approvals.

PROJECT TEAM RESPONSIBILITIES

- **Product Owner:** Rikita is the Product Owner. She will maintain the overall vision of the app, prioritize features, gather user feedback, and ensure the product aligns with user needs throughout the project.
- **Scrum Master:** Saina will serve as the initial Scrum Master. She will facilitate stand-ups, sprint planning, and retrospectives, remove blockers, and help the team follow agile best practices. The role may rotate as needed.
- **Frontend Developers:** Saul, Utshah, and Rakshya are responsible for frontend development. They will build the user interface using React Native, ensuring it is clean, ADHD-friendly, and accessible. Their work includes implementing barcode scanning, reminders, food tracking, and visual alerts that are intuitive for users with ADHD.
- **Backend Developers:** Mohammad is handling backend development. He will design and maintain the server, database, and APIs, manage user authentication, store food inventory, and schedule reminders. Their focus is on ensuring data reliability, security, and performance.
- **AI/ML Engineer:** Mohammad and Saina will support AI/ML development. They will integrate features like meal suggestions and personalized nudges using models such as GPT-4 or similar, based on food data and user behavior.
- **UI/UX Designer:** Utshah is designing ADHD-accessible interfaces and visual elements. She will conduct usability testing to ensure the app is intuitive and supportive for neurodivergent users.
- **QA Tester:** Rikita and Saul will handle testing. They will identify bugs, ensure accessibility, and verify that all features meet requirements and function properly for ADHD users.
- **Content Strategist/Writer:** Rakshya and Saul will create in-app messages, reminders, and tooltips that are simple, supportive, and easy to understand for users with ADHD.

TEAM STRUCTURE & COMMUNICATION

- The **Product Owner** role (Rikita) will stay fixed for consistency.
- The **Scrum Master** role (initially Saina) may rotate.
- Weekly sponsor meetings and daily team stand-ups will be held.
- Sprint reviews and retrospectives will be conducted to ensure continuous improvement.

9 COST PROPOSAL

This project is primarily software-based, so we do not anticipate significant expenses for hardware or fabrication. The main costs will likely involve software licenses or API usage, particularly for integrating AI features such as meal suggestions. We may also require a cloud backend service or database platform, but free or student-tier services will be used wherever possible. The full budget will be finalized by the end of Sprint 3, once development tools and technical requirements are confirmed.

9.1 PRELIMINARY BUDGET

At this stage, we expect potential expenses to include API access (such as OpenAI for meal and routine suggestions), backend services like Firebase or Supabase for data storage and user authentication, and a domain or hosting service if we deploy the app publicly. UI/UX design will use tools such as Figma, which are available under free student licenses. Additional tools like GitHub, Notion, or Trello for project management and collaboration are also free to use. Since we are building a mobile-first app using open-source technologies like React Native and Node.js, most development software will incur no cost.

9.2 CURRENT & PENDING SUPPORT

Currently, the CSE department provides default funding and access to development resources. We have access to free student licenses for most required tools. If additional funding is needed for API credits or hosting services, we will explore options through departmental support or external sponsors. These funding needs will be reassessed as the project progresses.

10 FACILITIES & EQUIPMENT

Since MindMeal is a software-focused project, we do not require any specialized lab equipment or hardware. Development and testing will primarily be done using personal laptops, UTA-provided machines, or in CSE computer labs as needed. For collaboration and team meetings, we plan to use available university facilities and Makerspace resources if necessary. All required development and design tools are accessible either for free or through student licenses, so no major equipment purchases are expected. Should any specific tools be needed later in the development cycle, we will plan to borrow or request access through the university.

11 ASSUMPTIONS

The following list contains critical assumptions related to the implementation and testing of the project.

- Users will be willing to initially invest time in scanning their pantry/products items.
- AI meal recommendation will maintain consistent accuracy and availability.
- The food database will provide accurate nutritional and allergen information.
- Users will interact with push notifications without becoming overwhelmed.
- The user will grant permissions for the camera and notifications.
- The database will be able to maintain all the necessary information.

12 CONSTRAINTS

The following list contains key constraints related to the implementation and testing of the project.

- The final prototype demonstration must be completed by Dec. 2025.
- The app must comply with federal regulations such as HIPAA.
- API calls services are limited.
- Limited budget for cloud services.
- The total development costs must not exceed \$800.

- Must work in IOS and Android platforms.
- Team members varying schedule due to other coursework or jobs.
- Cannot store personal health information without proper security measures.

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure.

Risk description	Probability	Loss (days)	Exposure (days)
Compatibility issues between IOS and Android	0.40	15	6
AI/APIs dependency	0.30	15	4.5
User data privacy compliance becomes more complex	0.40	10	4
App becomes more complex less user friendly	0.50	20	10
AI integration takes longer to integrate	0.30	10	3

Table 1: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

These deliverables are major grade components of the course. Completing these documents should generally be the sprint goal during the applicable sprint period. Refer to the current and previous course syllabus and schedules to estimate the due dates of these items. Remove this explanatory paragraph from your draft, but leave the heading.

14.1.1 PROJECT CHARTER

The project charter will be maintained collaboratively by the members of our project team throughout the development timeline. Updates will be made:

- At the end of each sprint to reflect any changes in project scope, objectives, deliverables, or risks.
- Whenever there is a significant change to the project plan, such as a change in team structure, timeline, or stakeholder expectations.
- Reviewed and approved by the team leader before finalizing updates.

All versions will be tracked in a version-controlled repository (GitHub) to ensure transparency and accountability.

- Initial Version Delivery: The initial version of the Project Charter will be delivered at the end of Sprint 1, July 11, 2025.
- Final Version Delivery: The final version will be delivered at the end of the project during the final sprint, incorporating any necessary updates made during the project.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification (SRS) document for the ADHD Food App will be maintained as a living document and updated regularly throughout the project lifecycle to ensure alignment with evolving user needs and project scope. It will be reviewed at the end of each sprint and revised immediately upon any significant changes to functionality, non-functional requirements, or technical constraints. All updates will be version controlled and stored in a shared repository to maintain transparency and traceability.

- **Initial Version Delivery:** The initial version of the SRS will be delivered at the end of Sprint 2, July 2025 once the core requirements have been defined.
- **Final Version Delivery:** The final version will be delivered at the end of the project during the final sprint, incorporating any necessary updates made during the project.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification for the ADHD Food App will be treated as a flexible working document that evolves with the project. We will review and update it at the end of each sprint, or sooner if there are major changes to the structure of the system, such as a change in technology, new components, or integration updates. It will be kept in a shared space so that the entire team can access the latest version at any time.

- **Initial Version Delivery:** The initial version will be completed once we have mapped out the core architecture and made key design decisions, likely by the end of Sprint 3.
- **Final Version Delivery:** The final version will be wrapped up before we move into final testing and deployment, making sure it reflects everything we actually built.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification for the ADHD Food App will be maintained as a dynamic document and updated throughout the development lifecycle to ensure it accurately reflects the system's evolving design. Updates will be made at the end of each sprint and as needed when significant changes occur to the application's features, user interface, data structures, or system components. All revisions will be documented and version-controlled within a shared repository to ensure consistency and accessibility across the team.

- **Initial Version Delivery:** The initial version of this document will be delivered at the beginning of detailed implementation planning, anticipated by the end of Sprint 4.
- **Final Version Delivery:** The final version will be completed prior to the final testing phase to ensure it accurately represents the implemented system design.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Items will be added to the product backlog by the Product Owner (Rikita) based on SRS requirements, user feedback, and group vote. Priority will be determined using the MoSCoW method. We will use GitHub for backlog management.

14.2.2 SPRINT PLANNING

We plan to run two-week sprints across two semesters, totaling nine sprints from May through December 2025. This total includes four sprints outlined in the syllabus and five additional sprints based on the Sprint 25 syllabus.

14.2.3 SPRINT GOAL

The sprint goal will be determined during a group meeting, based on the next set of prioritized requirements outlined in the Software Requirements Specification (SRS) document.

14.2.4 SPRINT BACKLOG

The team will collaboratively decide which product backlog items are moved into the sprint backlog during sprint planning meetings. The sprint backlog will be maintained and updated using GitHub Projects, which will serve as our collaborative scrum board for tracking tasks, progress, and priorities.

14.2.5 TASK BREAKDOWN

Tasks will be assigned based on each team member's strengths and the roles outlined in the Roles & Responsibilities section. Team members are encouraged to collaborate and support each other, especially if they complete their assigned tasks early. Task ownership and workload distribution will be discussed during sprint planning to ensure alignment and accountability.

14.2.6 SPRINT BURN DOWN CHARTS

The Scrum Master will be responsible for generating the burn down chart for each sprint. To track effort, team members will log their completed tasks on GitHub and update their estimated hours in a shared Excel sheet. This data will be reviewed regularly and used to update the burn down chart.

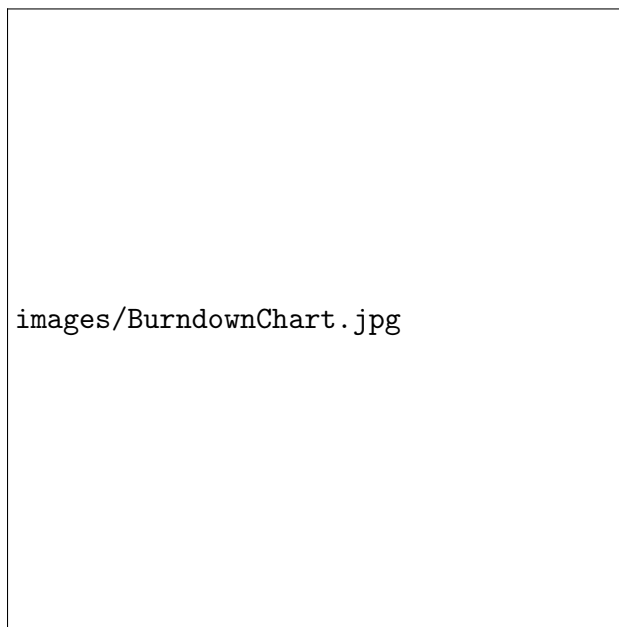


Figure 1: Example of burn down chart

14.2.7 SPRINT RETROSPECTIVE

Sprint retrospectives will be held at the end of each sprint during the final team meeting. The goal is to reflect what went well, identify future challenges, and improvements in the next sprint.

14.2.8 INDIVIDUAL STATUS REPORTS

Each member will submit a bi-weekly report detailing:

- Task completed
- Task in progress
- Issues encountered
- Hours spent
- Future tasks

The reports will be submitted and reviewed by the group members to identify any issues and ensure progress.

14.2.9 ENGINEERING NOTEBOOKS

Each member will update their Engineering Notebook at least once per week. Entries must include sketches, task, meeting notes, etc.

14.3 CLOSEOUT MATERIALS

At the end of the project, the team will compile and submit the following closeout material:

- Final presentation slides
- Final report
- Recorded demo
- Documentation and instructions

The material will be submitted by the designated deadline.

14.3.1 SYSTEM PROTOTYPE

The final prototype will consist of a fully functional mobile application that includes bar-code scanning, AI-powered meal recommendations based on user conditions, and a notification system to track eating habits. Core features will be demonstrated during the seventh sprint, with the final system demonstration scheduled for December.

14.3.2 PROJECT POSTER

The project poster will provide a clear and engaging overview of the MindMeal app, including a brief explanation of the challenges ADHD individuals face with food management. It will showcase key app features such as inventory logging, expiration notifications, and meal suggestions, along with mockups of the user interface. The poster will also highlight how MindMeal is designed specifically for ADHD users, emphasizing simplicity and accessibility. With a focus on food waste reduction and healthier eating habits, the poster will use large fonts, simple icons, and a cohesive color scheme. It will be printed in a standard 36"x24" format and will be ready by the end of October 2025, with updates reflecting project progress.

14.3.3 WEB PAGE

What will be included on the project web page? Will it be accessible to the public? When will this be delivered? Will it be updated throughout the project, or just provided at closeout (at a minimum, you need to provide a simple web page at the end).

14.3.4 DEMO VIDEO

The explanatory video will go through how MindMeal works, explaining what it actually is, and why it is significant. The video starts with a brief description of the problem — which is that people with ADHD tend to forget that they have food at home, and hence one of the results is food getting expired, money getting wasted, or meals missed. Next, we will move onto a real-time view of the app.

The video will show these steps: one: the user puts food into inventory using either a photo or a barcode; two, checking on food about to expire; and three, receiving gentle reminders throughout the day. One segment will be dedicated to demonstrating how the AI assistant suggests a recipe based on what the user actually has inside their fridge. The idea behind this will be to keep things simple, helpful, and tailored for someone that maybe does not have a solid hold on routines.

The team is also looking into including a couple of voices behind the scenes, from app designing and coding to its testing, to make it feel lighter and more relatable. This additional footage will also come in handy for future edits or presentations.

The full video will be around 3 to 4 minutes long. It will focus on the core features, the user experience, and the problem we're trying to solve, in a way that's clear for anyone watching whether that's our professor, classmates, or someone who might actually benefit from using the app.

14.3.5 SOURCE CODE

The source code for MindMeal will be maintained using Git and hosted on GitHub to track changes and facilitate collaboration. The customer will receive the complete source code at project completion, either via GitHub repository or a ZIP file. After project closure, MindMeal will be open-sourced under the MIT License, allowing free use, modification, and distribution. License terms will be included in the LICENSE file and the project's README.

14.3.6 SOURCE CODE DOCUMENTATION

MindMeal's documentation will follow industry-standard practices to ensure clarity and accessibility. We will use Doxygen to generate detailed code documentation, including API references and inline comments. The final documentation will include an overview of the project, setup instructions, user guides, and detailed technical documentation for developers. The documentation will be provided in browsable HTML format for ease of navigation and also in PDF format for offline access. This approach will ensure that both technical and non-technical stakeholders can easily understand and use the app.

14.3.7 HARDWARE SCHEMATICS

No hardware for the project.

14.3.8 CAD FILES

No CAD files for the project.

14.3.9 INSTALLATION SCRIPTS

For the ADHD Food App, the software will be distributed through official mobile app stores (Google Play and Apple App Store), so users will not need to run installation scripts manually. For development and deployment purposes, we will provide installation and setup scripts for internal use, including separate scripts or configuration files for the front-end and back-end components. These may include environment setup, dependency installation, and database initialization tools to streamline deployment for testing or future scalability. No manual installation will be required for end users.

14.3.10 USER MANUAL

The ADHD Food App will include a digital user manual, accessible within the app and as a downloadable PDF. It will provide clear, step-by-step instructions tailored to our users' needs. A printed manual or setup

video will not be provided, as digital documentation is sufficient and easier to maintain.

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[1] [2] [3]